

Continuous Replenishment System (CRS)

AI-Powered Vendor-Managed Inventory for FMCG Distribution

FMCG Beverages · Australia

TFT · SBA · OR-Tools MILP

↗ Temporal.io Saga Orchestration

5 Distributors · 30 SKUs · 52-Week History

Live Demo Available

DIRECTLY ADDRESSES CLIENT ASSESSMENT TOPICS

This document and live demo answer all three benchmark questions:

- (1) Forecast & replenishment generation at distributor-SKU level
- (2) Override & modification workflow with full audit trail
- (3) Fair-share credit allocation using MILP optimisation

Prepared by Abhishek Sharma · Autonomous VMI Architecture · azure-01 Fleet Node

SECTION 1

What is CRS?

An autonomous, end-to-end replenishment engine that eliminates manual spreadsheet-driven ordering in FMCG distribution networks.

POS Telemetry Ingestion

Real-time distributor sell-out data, stockout detection, and Bayesian imputation of missing periods

Algorithmic Demand Forecasting

ADI/CV² routing to ETS, SBA, ARIMA, or TFT — per SKU, per distributor, every cycle

ROQ Generation

Target Stock Level (TSL) → Replenishment Order Quantity with MOQ, case-pack, pallet, and DOS ceiling constraints

Credit-Aware Allocation

OR-Tools MILP maximises minimum Days-of-Stock across distributors given live AR/credit limits



Temporal.io Saga Orchestration

Durable, compensating workflow handles approvals, ERP release, and rollback atomically

4-Persona Dashboards

Demand Planner · Sales Executive · Distributor · Credit Admin — purpose-built UX per role

REPLENISHMENT CYCLE

End-to-End Flow



Bullwhip Effect Prevention: CRS operates on distributor *sell-out* POS data (not distributor purchase orders), eliminating the demand signal amplification that causes the bullwhip effect across supply chain echelons. Stockout periods are Bayesian-imputed using surrounding demand patterns rather than treated as zero demand.

SECTION 2 — ASSESSMENT Q&A

Client Benchmark Topics Answered

Technical depth for each of the three assessment questions from the consulting brief.

TOPIC 1 · FORECAST & REPLENISHMENT GENERATION

How does CRS generate forecasts and replenishment recommendations at the distributor-SKU level?

Step 1 — Demand Classification (ADI/CV² Matrix):

- ADI (Average Inter-Demand Interval) & CV² (Squared CoV) computed per SKU
- $ADI < 1.32 + CV^2 < 0.49 \rightarrow$ **ETS** (smooth, fast-moving)
- $ADI \geq 1.32 + CV^2 < 0.49 \rightarrow$ **SBA** (intermittent)
- $ADI < 1.32 + CV^2 \geq 0.49 \rightarrow$ **ARIMA** (erratic)
- $ADI \geq 1.32 + CV^2 \geq 0.49 \rightarrow$ **SBA** (lumpy)
- Revenue-A or high covariate $R^2 \rightarrow$ **TFT** (neural, multi-horizon)

Step 2 — Syntetos-Boylan Approximation (SBA) for intermittent/lumpy:

$$f_{SBA} = (1 - \alpha/2) \times \bar{z}_t / \bar{p}_t$$

Debiases Croston's method by the factor $(1-\alpha/2)$ where α is the smoothing constant, $\bar{z}_t$ is the demand-size estimate, and $\bar{p}_t$ is the inter-demand interval estimate.

Step 3 — Target Stock Level (TSL):

$$TSL = \bar{D}(T+L) + Z \cdot \sqrt{(L \cdot \sigma_D^2 + \bar{D}^2 \cdot \sigma_L^2)}$$

T=review period, L=lead time, σ_D =demand std dev, σ_L =lead-time std dev, Z=service-level z-score. Includes lead-time variance decomposition.

Step 4 — ROQ with logistics constraints:

- $ROQ_{raw} = \max(0, TSL - \text{Net Inventory Position})$
- Apply MOQ floor $\rightarrow$ round up to case pack $\rightarrow$ pallet heuristic (round up if remainder $\geq 40\%$ of pallet)
- DOS ceiling cap: if projected DOS > 90 days $\rightarrow$ hard cut to prevent hoarding

TOPIC 2 · OVERRIDE & MODIFICATION WORKFLOW

At what stages can Sales, planners, or distributors override CRS-generated recommendations?

4 Override Entry Points:

- **Demand Planner:** Adjusts forecast quantity in Exception Triage panel (P90 breach, negative FVA, model reclassification). Override range gated to $\pm 15\%$ of P50 to prevent arbitrary changes.
- **Sales / Executive:** Revenue-target simulator adjusts network-level uplift applied proportionally across distributors using revenue tier weights.
- **Distributor:** Requests quantity adjustments via portal (e.g., pre-promotional build-up, warehouse capacity constraints). Triggers re-approval workflow.
- **Credit Admin:** Manual credit-hold release unblocks distributors from CREDIT_HOLD status, allowing solver to process their allocations.

Override Accountability (Forecast Value Added — FVA):

FVA = MAPE_naive - MAPE_adjusted

Every manual override is logged with original_qty, overridden_qty, pct_change, reason_code, user_id, and FVA scores in audit_override_log. Negative FVA (override made forecast worse than the naive baseline) triggers an exception flag in the next cycle, visible to the Demand Planner.

Reason Codes tracked:

- PROMOTIONAL_UPLIFT · COMPETITOR_DISRUPTION
- SEASONAL_PRE_BUILD · ERRONEOUS_FORECAST
- WAREHOUSE_CAPACITY · CREDIT_CONSTRAINT

Temporal.io Human-in-the-Loop Gate: After ROQ calculation and before ERP release, the saga pauses at APPROVAL_WAIT. A senior planner can approve or reject via API signal. Rejection triggers compensating transactions to reverse any ERP credit holds already placed.

TOPIC 3 · FAIR-SHARE CREDIT ALLOCATION

When a distributor's credit is insufficient, how does CRS allocate the available amount across SKUs?

OR-Tools MILP Formulation (SCIP backend, <2ms solve time):

Maximise: z_min (minimum DOS across all eligible SKUs)

Subject to: $\sum (x_i \times \text{unit_cost}_i) \leq \text{credit_available}$

$\forall i: x_i \times \text{demand}_i - z_{\min} \times \text{demand}_i \geq -\text{NIP}_i \text{ (DOS} \geq z_{\min}\text{)}$

$\forall i \text{ Tier-1: } x_i \geq 0.90 \times \text{ROQ}_i \text{ (90\% floor reservation)}$

Where x_i = allocated order quantity, $z_{\min}$ = auxiliary continuous variable = minimum DOS any SKU achieves.

Tier Prioritisation:

- Tier 1 (A-class revenue):** 90% floor reservation — never cut below 90% of their ROQ regardless of credit pressure
- Tier 2 (B-class):** Participate in fair-share allocation pool
- Tier 3 (C-class):** Excluded from allocation when credit < 60% of total ROQ value

What "fair share" means: Rather than first-come-first-served or proportional-cut, the MILP finds the allocation that maximises the worst-off distributor's DOS. If two distributors have different demand rates, they receive different quantities but identical Days-of-Stock coverage — genuinely equitable, not just proportional.

Output logged to: credit_allocation_ledger with solver_status (WITHIN_LIMIT / CREDIT_HOLD / FAIR_SHARE_APPLIED), solver_run_ms, and allocated_order_val per distributor-SKU batch.

Persona 1 — Demand Planner

Exception-driven workflow · Forecast explainability · Human override with FVA accountability

The Demand Planner dashboard surfaces only items that need human attention — outliers, reclassifications, and negative FVA overrides — rather than showing all SKUs. This exception-driven UX is critical for planners managing thousands of SKU-distributor combinations.

Exception Triage Grid

Automatically flags SKUs requiring planner attention:

- P90 BREACH** Actuals exceeded 90th-percentile forecast band — model underestimating demand
- NEGATIVE FVA** Previous manual override made accuracy worse than naive baseline
- RECLASSIFICATION** Demand pattern changed class (e.g., ETS→TFT) this cycle

P10/P50/P90 Forecast Bands

Probabilistic forecast bands from the TFT/ETS/SBA model:

- P10** — 10th percentile (pessimistic)
- P50** — Median forecast (planning baseline)
- P90** — 90th percentile (safety stock trigger)
- Actual demand overlay shows historical accuracy

TFT Attention Weights

Model interpretability for the TFT neural forecaster — shows which input features most influenced this cycle's forecast:

- Lag Sales W-52 (31%) · Promo Depth (24%)
- Day of Week (18%) · Weather Index (12%)
- Lag Sales W-4 (8%) · Holiday Flag (5%)

Allows planners to challenge the model's reasoning, not just its number.

Override Submission

Inline override form per SKU:

- Quantity slider gated to ±15% of P50
- Mandatory reason code selection
- Submitted override triggers FVA re-evaluation next cycle
- Full audit trail in audit_override_log

CRS · Demand Planner DashboardWeek 34, 2026 · Cycle: Auto-generated

SKU	EXCEPTION	MODEL	P50 FORECAST	FVA SCORE	ACTION
Solo Lemon 375mL Can 24pk	P90 BREACH	TFT	452 units	+1.8 pp	Override ↗
Powerade Mountain Blast 600mL	NEGATIVE FVA	ETS	218 units	−6.2 pp	Override ↗
Golden Circle OJ 2L 6pk	RECLASSIFICATION	ETS→TFT	891 units	—	Review ↗
V Energy 250mL 24pk	P90 BREACH	SBA	134 units	+2.1 pp	Override ↗
Bundaberg Ginger Beer 375mL	NEGATIVE FVA	ARIMA	67 units	−3.4 pp	Override ↗

P10/P50/P90 Forecast Bands — Solo Lemon 375mL (14-Day Horizon)

TFT Attention Weights — Feature Importance

Persona 2 — Sales / Executive

Network KPI health · S&OP revenue simulator · DOS by distributor · Strategic levers

The Executive dashboard aggregates the replenishment network into a single-glance view — are we on track with revenue targets, which distributors are approaching stockout, and what happens to revenue if we adjust the service level policy?

Network KPI Dashboard

- **Average Network DOS** — weighted by distributor revenue; 28.4 days target
- **Stockout Risk SKUs** — count of SKU-distributor pairs below safety stock
- **Credit Headroom %** — available credit as % of recommended order value
- **Cycle Fill Rate** — % of recommended ROQ that cleared credit and ERP

S&OP Revenue Simulator

Interactive slider to model revenue impact of policy changes:

- Adjust target DOS from 14 to 60 days
- See projected revenue uplift vs. working capital cost
- Model promotional uplift % across all distributors
- Output feeds directly into next cycle's ROQ calculation

📊 Distributor DOS Health Map

- Bar chart of current DOS by distributor
- Red threshold line at safety stock equivalent
- DIST-AU-BNE highlighted in CREDIT_HOLD status

- Colour coding: green ≥ 21 days, amber 14-21, red < 14

 Credit Hold Impact View

- Shows revenue at risk from distributors in CREDIT_HOLD
- Links to Credit Admin for one-click release approval
- Historical credit utilisation trend (30-day rolling)

AVG NETWORK DOS

28.4

↑ 2.1 vs last cycle

STOCKOUT RISK SKUS

7

↑ 3 new this cycle

CYCLE FILL RATE

94.2%

↑ 0.8pp vs target

REVENUE AT RISK

\$182K

DIST-AU-BNE hold

Days-of-Stock by Distributor — Current Cycle

Persona 3 — Distributor Portal

In-transit tracking · 14-day DOS projection · Adjustment requests · Self-service visibility

Distributors traditionally operate with zero visibility into the replenishment logic that governs their stock. The Distributor Portal gives them a self-service view: what's inbound, when it arrives, and projected stock health by SKU — with a structured channel to request adjustments.

In-Transit Shipment Tracker

- All open replenishment orders with ETA and quantity
- Status: CONFIRMED → IN_TRANSIT → DELIVERED
- Alerts for shipments > 2 days past expected ETA
- Consolidated view across all SKUs in one distributor

14-Day DOS Projection

- Area chart showing projected stock level for top SKUs
- Inbound inventory incorporated into projection curve
- Red threshold line at service-level safety stock
- Helps distributor plan warehouse capacity and promotions

Adjustment Request Form

- Structured form: SKU, requested quantity, reason, priority
- Reason codes: PROMO, CAPACITY, COMPETITIVE, OTHER
- Triggers Temporal.io workflow — planner gets notified immediately
- Distributor sees request status in real-time

Credit Utilisation View

- Shows current AR balance vs. credit limit
- Utilisation % with colour-coded status (green/amber/red)
- Estimated days until next payment clearing
- CREDIT_HOLD notification with escalation path

ORDER ID	SKU	QTY	STATUS	ETA	VALUE
ORD-2026-0841	Solo Lemon 375mL 24pk	480 units	IN_TRANSIT	23 Aug	\$3,840
ORD-2026-0842	Powerade Mountain Blast	240 units	IN_TRANSIT	23 Aug	\$1,680
ORD-2026-0843	Golden Circle OJ 2L	96 units	CONFIRMED	25 Aug	\$2,112
ORD-2026-0839	Bundaberg Ginger Beer	144 units	⚠ ETA DELAYED	+2d	\$1,584

14-Day Projected DOS — Top 3 SKUs for DIST-AU-BNE

Persona 4 — Credit Admin

Credit utilisation monitoring · OR-Tools MILP output · Manual credit release · Ledger audit

The Credit Admin has a single critical responsibility: ensure that replenishment doesn't violate credit risk policy while also not unnecessarily blocking legitimate orders. They see the MILP solver output directly — why certain SKUs were cut, what the fair-share allocation was, and have a one-click release mechanism for credit holds.

- ### Credit Utilisation Table

 - All distributors with live AR balance, credit limit, utilisation %
 - DIST-AU-BNE at 80% = **CREDIT_HOLD**
 - Threshold alerts: >75% = amber, >90% = red, 100% = hold
 - 30-day trend sparkline per distributor
- ### OR-Tools MILP Solver Output

 - Shows solver status: OPTIMAL / INFEASIBLE / FAIR_SHARE_APPLIED
 - Per-SKU: recommended ROQ vs. allocated quantity, DOS impact
 - Solve time (typically < 5ms for 30-SKU batch)
 - z_min value = minimum DOS the allocation guarantees any SKU
- ### Manual Credit Release

 - One-click release for CREDIT_HOLD distributors
 - Confirmation modal: shows amount being unblocked + risk
 - Release logged to credit_allocation_ledger as MANUAL_RELEASE
 - Triggers immediate workflow re-run for that distributor
- ### Allocation Ledger

 - Full history: solver_status, credit_available, proposed vs. allocated order value
 - Audit trail for finance/compliance teams
 - GENERATED column: credit_available = credit_total – AR – unbilled
 - Downloadable CSV for reconciliation

CRS · Credit Admin Dashboard

DISTRIBUTORS ON HOLD

1

DIST-AU-BNE

TOTAL CREDIT AVAILABLE

\$2.14M

74% headroom

MILP LAST SOLVE

OPTIMAL

2ms · 3 SKUs cut

Credit Utilisation by Distributor

DISTRIBUTOR	CREDIT LIMIT	AR OUTSTANDING	AVAILABLE	UTILISATION	STATUS
DIST-AU-SYD	\$500K	\$142K	\$358K	28.4%	ACTIVE
DIST-AU-MEL	\$750K	\$310K	\$440K	41.3%	ACTIVE
DIST-AU-BNE	\$300K	\$241K	\$59K	80.3%	CREDIT_HOLD
DIST-AU-PER	\$400K	\$198K	\$202K	49.5%	ACTIVE
DIST-AU-ADL	\$250K	\$87K	\$163K	34.8%	ACTIVE

OR-Tools MILP — How the Solver Works

```
# Google OR-Tools SCIP backend — Fair-Share Allocation
solver = cp_model.CpModel() # MILP, not CP — uses SCIP

# Objective: maximise the minimum DOS any SKU achieves
z_min = solver.NewIntVar(0, 90, 'z_min') # DOS ceiling = 90 days

for sku in tier_1_skus:
    # Tier-1 floor: allocate ≥ 90% of recommended ROQ
    solver.Add(x[sku] >= 0.90 * sku.roq)

for sku in all_eligible_skus:
    # DOS constraint: allocation + NIP must achieve at least z_min DOS
    solver.Add(x[sku] * sku.demand - z_min * sku.demand >= -sku.nip)

# Budget constraint: total order value ≤ credit available
solver.Add(sum(x[s] * s.unit_cost for s in skus) <= credit_available)

solver.Maximize(z_min)
```

SECTION 7 — ARCHITECTURE

Technical Stack & Design Decisions

Production-grade components, demo-scale configuration — the same algorithms would power a live VMI deployment.

➤ Temporal.io Saga — 6-Stage Durable Workflow



Compensating Transaction: If ERP order creation fails after credit hold was placed, Temporal automatically executes `reverse_erp_credit_hold` to undo the hold — maintaining distributed consistency without a database transaction spanning multiple services.

📦 PostgreSQL 16

- Multi-tenant shared schema with `tenant_id`
- Partitioned fact tables (by year)
- Generated columns: `net_inventory_pos`, `pct_change`, `credit_available`
- RLS-ready for tenant isolation

➤ FastAPI + asyncpg

- OpenAPI 3.0 spec with Swagger UI at `/docs`
- Async connection pool (`asyncpg`)
- Persona-based JWT authentication
- `structlog` for structured JSON logging

Forecasting Engine

- Pure Python: `numpy` + `scipy` (no ML infra needed)
- SBA: Syntetos-Boylan debiased Croston
- ETS: Simple exponential smoothing + quantile bands
- TFT: Attention-based (architecture demo)

⚙️ React + Vite + Recharts

- TypeScript + Tailwind CSS
- Zustand for auth state (persisted)
- TanStack Query for API caching
- Recharts: `AreaChart`, `BarChart`, `ReferenceLine`

Temporal.io 1.24

- Auto-setup with `postgres12` backend
- Workflow signals for human approval gate
- Workflow queries for state inspection
- Retry policies per activity type

Synthetic FMCG Data

- 5 AU distributors × 30 beverage SKUs
- 52 weeks with realistic seasonality
- Modelled stockout events + Bayesian imputation
- ABC-XYZ segmentation (A/B/C revenue tiers)

Demo Environment & Access

All services are running live. No installation required — access directly from any browser.

Live Demo — Running on Azure Cloud

Full Docker Compose stack: PostgreSQL · Temporal.io · FastAPI · React Frontend

FRONTEND DASHBOARD

http://172.174.209.56:3010

4 persona buttons · No login required

API DOCUMENTATION

http://172.174.209.56:8001/docs

Swagger UI · All endpoints live

TEMPORAL WORKFLOW UI

http://172.174.209.56:8081

Watch saga execution in real-time

SOURCE CODE (PRIVATE)

github.com/abhiavi/crs-demo

Request collaborator access

Demo Walkthrough — 5-Minute Script

Step 1 — Demand Planner (90s)

- Login as Demand Planner
- Review Exception Triage grid — note P90 BREACH on Solo Lemon
- Expand TFT attention weights — explain "promotional depth" driving 24% of forecast
- Submit an override with reason PROMOTIONAL_UPLIFT

Step 2 — Credit Admin (60s)

- Switch to Credit Admin persona
- Show DIST-AU-BNE in CREDIT_HOLD at 80% utilisation
- Click "View MILP Solver Output" — show z_min = 18.3 days DOS guaranteed
- Click "Release Credit Hold" and confirm

Step 3 — Trigger Workflow (60s)

- Hit POST /api/v1/workflows/trigger in Swagger UI
- Switch to Temporal UI — watch saga progress through stages
- Show the APPROVAL_WAIT pause (human-in-the-loop gate)
- Send approval signal to complete the workflow

Step 4 — Distributor (30s)

- Switch to Distributor persona
- Show in-transit orders for DIST-AU-BNE
- Highlight 14-day DOS projection curve

- Show adjustment request form

Key Talking Points for Each Client Question:

Q1 (Forecast generation): "The system classifies every SKU-distributor pair on the ADI/CV² matrix each week. High-value SKUs get the TFT model with probabilistic P10/P50/P90 bands. Intermittent-demand SKUs use SBA, which debiases the classic Croston method by a factor of $(1-\alpha/2)$. The Target Stock Level formula explicitly accounts for lead-time variance, not just demand variance."

Q2 (Overrides): "Any of the 4 personas can influence the replenishment — but every change is logged with the original quantity, the new quantity, the reason code, and the user ID. More importantly, the system calculates Forecast Value Added: if your override made the forecast worse than a naive baseline, you get flagged on the next cycle. This creates accountability without restricting human judgment."

Q3 (Fair-share credit): "We use a MILP solver from Google OR-Tools. The objective is to maximise the minimum Days-of-Stock any SKU achieves — not to cut proportionally. Tier-1 SKUs get a 90% floor, so strategic products are never starved. The solver runs in under 5 milliseconds for a 30-SKU batch, so it fits inside the replenishment cycle runtime."

Continuous Replenishment System (CRS) · FMCG Beverages Demo · Australia

Built with Temporal.io · FastAPI · OR-Tools MILP · React · PostgreSQL 16 · Deployed on Azure

Confidential — Prepared for APAC Consulting Engagement · August 2026